

Financial Time Series Analysis

Lecture 2

- ▶ Welcome class of 2022 MSCF admitted students.
- ▶ TA Session, Friday, March 26, 11:00am-12:00pm (EDT)

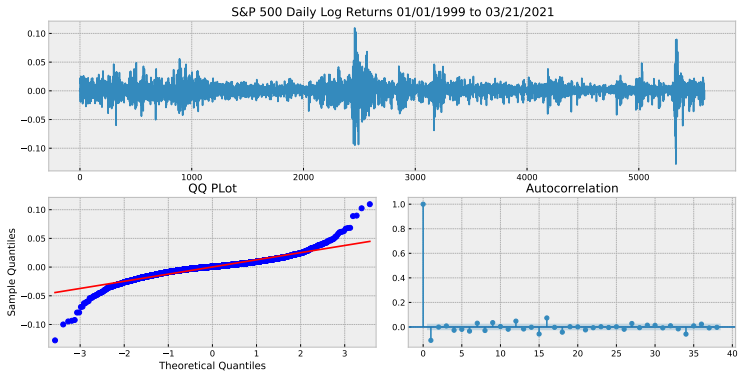
Outline

- ▶ Review
- ▶ Stationarity
 - ▶ Strong/Weak/White Noise
 - ▶ Martingale Difference Sequences (MDS)
 - ▶ Wold Decomposition
- ▶ Standard Time Series Models: $AR(p)$, $MA(q)$, $ARMA(p,q)$

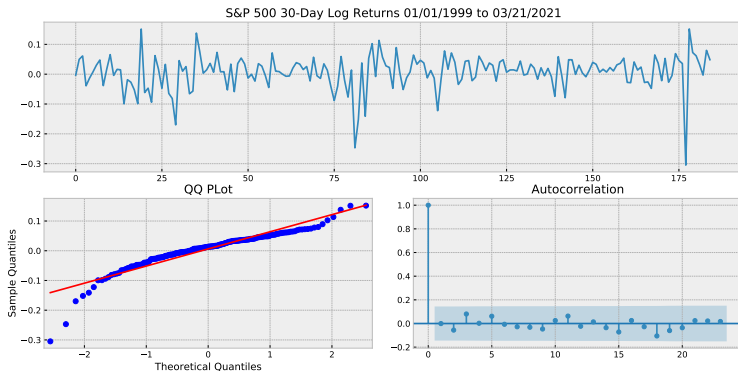
Brief Review

- ▶ Eleven stylized facts about asset prices
 - ▶ Daily log-returns have heavier tails than the normal distribution
 - ▶ As time interval increases log-returns become more normal
 - ▶ Little autocorrelation at daily time scale
 - ▶ Gains/loss asymmetry (larger drawdowns than upward moves)
 - ▶ Volatility clustering
 - ▶ Leverage effect, volatility is negatively correlated with returns.

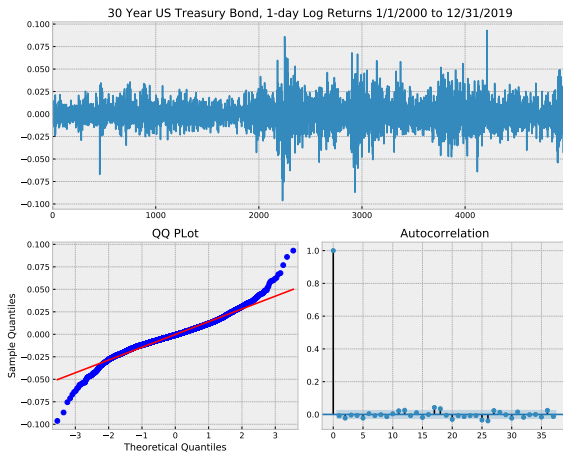
S&P 500 Daily Log Returns 1/1/1999-3/20/2021



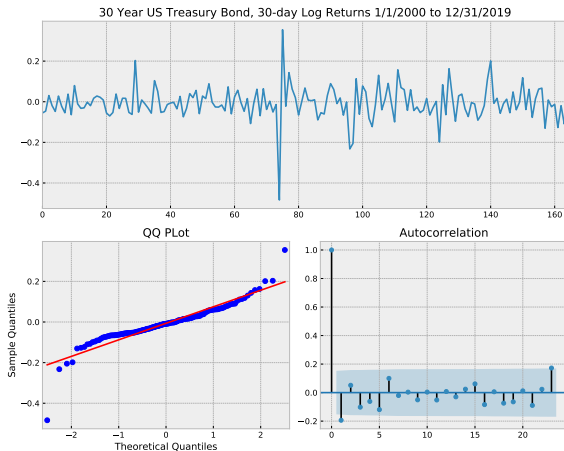
S&P 500 30-Day Log Returns 1/1/1999-3/20/2021



US 30-Year Treasury Bond Daily Returns 1/1/2000-12/31/2019



US 30-Year Treasury Bond 30-Day Returns 1/1/2000-12/31/2019



STATIONARITY

Stationarity: 1

- ▶ Consider a time series $\{X_t, -\infty < t < \infty\}$ or $\{X_t, 0 < t < \infty\}$. This time series is called
 - ▶ **Strictly Stationary** if for all $n \geq 1$, $t_1 < t_2 < \dots < t_n$, and $i = 1, 2, \dots$ the joint distribution of $(X_{t_1}, X_{t_2}, \dots, X_{t_n})$ is the same as $(X_{t_1+i}, X_{t_2+i}, \dots, X_{t_n+i})$.
 - ▶ **Weakly Stationary** if for all $t_1 < t_2$ and $i = 1, 2, \dots$, $\text{Cov}(X_{t_1}, X_{t_2}) = \text{Cov}(X_{t_1+i}, X_{t_2+i})$.
- ▶ Weak stationarity and jointly normal distributions imply strict stationarity.
- ▶ We will always use weak stationarity, which is also known as covariance stationarity, second-order stationarity, and wide-sense stationarity.

Stationarity: 2

- ▶ A weakly stationary process $\{X_t, t \in \mathcal{Z}\}$ satisfies:
 - ▶ $E(X_t) = \mu, t \in \mathcal{Z},$
 - ▶ $Cov(X_s, X_t) = \gamma(s, t) = \gamma(s + k, t + k), s, t, k \in \mathcal{Z}$
- ▶ Since $\gamma(s, t) = \gamma(0, t - s) = \gamma(t - s, 0)$, the function γ is actually a function of only one variable, $|t - s|$ is the **lag**.
- ▶ We define the **autocovariance function**, $\gamma(h) = \gamma(h, 0)$
- ▶ Note that $\gamma(0) = Var(X_t)$ and the autocorrelation function, $\rho(h)$, is defined by $\rho(h) = \gamma(h)/\gamma(0)$.

Stationarity: 3

- ▶ The backbone of much of basic statistics is the i.i.d. process. In time series analysis, the corresponding backbone is the **White Noise Process**
- ▶ A **white noise process** is a weakly stationary process with constant mean 0, and autocovariance function

$$\gamma(h) = \begin{cases} \sigma^2 & \text{if } h = 0, \\ 0 & \text{if } h \neq 0. \end{cases}$$

- ▶ A white noise process is denoted by $WN(0, \sigma^2)$.
- ▶ The autocorrelation function for a white noise process is,

$$\rho(h) = \begin{cases} 1 & \text{if } h = 0, \\ 0 & \text{if } h \neq 0. \end{cases}$$

Asymptotics of Stationary Sequences: 1

- ▶ While it is certainly possible to define stochastic sequences that are more general than i.i.d. sequences or independent sequences, we need to have sequences that have good asymptotic properties like laws of large numbers and central limit theorems.
- ▶ These theorems are important for having the consistency of statistical procedures, the asymptotic distributions of estimators of model parameters, the critical values for test statistics, or the width of prediction/forecast intervals being characterizable.
- ▶ We need the asymptotic behavior of our models to be characterizable.

Asymptotics of Stationary Sequences: 2

- ▶ Consider the example where X_0 is a Bernoulli(.5) random variable, and $X_n = X_0$ for all n . This results in two possible outcomes for the entire process: $(0, 0, 0, \dots)$ and $(1, 1, 1, \dots)$.
- ▶ This process is both strictly stationary and weakly stationary with $\gamma(h) = .25$, $\rho(h) = 1$, $\forall h \geq 0$.
- ▶ $S_n/n = (X_1 + \dots + X_n)/n = X_0$.
- ▶ This is an example in which the long run average is identical to X_0 almost surely, not the mean value of .5.
- ▶ Suppose X_t , $t \geq 0$ is weakly stationary with $E(X_t) = \mu$ and $\text{Var}(X_n) = \sigma^2$. If $\sum_{i=1}^{\infty} |\gamma(i)| < \infty$, then $\sum_{i=1}^n X_i/n \rightarrow \mu$ in probability.

Standard Facts on Conditional Expectation: 1

Consider a time series $\{X_t, -\infty < t < \infty\}$. We define $\mathcal{F}_t$ to be the σ -algebra generated by $(\cdots, X_{t-1}, X_t)$. Intuitively this represents “all the information in the random variables $(\cdots, X_{t-1}, X_t)$ ”. Important properties include

- ▶ $\mathcal{F}_s \subset \mathcal{F}_t$, for $s \leq t$
- ▶ $E(X|\mathcal{F}_s)$ is $\mathcal{F}_s$ measurable
- ▶ $E(Y) = E(E(Y|\mathcal{F}_t)) \forall t$ and Y with $E|Y| < \infty$.
- ▶ $E(X|\mathcal{F}_t) = X$ a.s. if X is $\mathcal{F}_t$ measurable
- ▶ $E(XY|\mathcal{F}_t) = YE(X|\mathcal{F}_t)$, a.s. if Y is $\mathcal{F}_t$ measurable.
- ▶ $E(E(X|\mathcal{F}_s)|\mathcal{F}_t) = E(E(X|\mathcal{F}_t)|\mathcal{F}_s) = E(X|\mathcal{F}_s)$ for $s < t$.

Martingale Difference Sequences: 1

- ▶ Recall that a **martingale sequence** is a pair $\{(X_t, \mathcal{F}_t), t \geq 0\}$ with $\mathcal{F}_s \subset \mathcal{F}_t, \forall s \leq t$ and $E|X_t| < \infty$, satisfying

$$E(X_{t+1}|\mathcal{F}_t) = X_t, a.s. \forall t \geq 0.$$

- ▶ In a similar fashion we define a **martingale difference sequence** (MDS) to be a pair $\{(Y_t, \mathcal{F}_t), t \geq 0\}$ with $\mathcal{F}_s \subset \mathcal{F}_t, \forall s \leq t$ and $E|Y_t| < \infty$, satisfying

$$E(Y_{t+1}|\mathcal{F}_t) = 0, a.s., \forall t \geq 0.$$

- ▶ By taking expectations, $E(Y_t) = 0$.
- ▶ Note that we can rewrite the martingale property as $E(X_{t+1} - X_t|\mathcal{F}_t) = 0, a.s. \forall t \geq 0$. If we let $Y_{t+1} = X_{t+1} - X_t$, then $\{Y_t, t \geq 0\}$ is an MDS.

Martingale Difference Sequences: 2

There are three different types of noise processes that we will be considering:

1. Independent sequences, $\{X_t, t \geq 0\}$ with common mean μ and variance σ^2 . Sometimes this is specialized to i.i.d. sequences and even further to i.i.d. with $N(\mu, \sigma^2)$ distribution.
 2. Martingale difference sequences (MDS):
 $E(X_{t+1}|\mathcal{F}_t) = 0, a.s.$
 3. Weakly stationary processes in which $E(X_t) = \mu$,
 $\text{Cov}(X_s, X_t) = \gamma(t - s)$.
- An independent sequence with $\mu = 0$ is clearly an MDS.

$$\{\text{Independent Sequences}\} \subset \{\text{MDS}\}.$$

Martingale Difference Sequences: 3

- ▶ MDS with common finite variance are white noise processes, therefore are weakly stationary, i.e.

$\{\text{Common Finite Variance MDS}\} \subset \{\text{White Noise Processes}\}.$

- ▶ If $\{X_t, t \geq 0\}$ is an MDS, and $s < t$

$$\begin{aligned}\text{Cov}(X_s, X_t) &= E(X_s X_t) - 0 \cdot 0 \\ &= E(E(X_s X_t | \mathcal{F}_{t-1})) \\ &= E(X_s E(X_t | \mathcal{F}_{t-1})) \\ &= E(X_s \cdot 0) \\ &= 0\end{aligned}$$

Martingale Difference Sequences: 4

- ▶ Why introduce this “intermediate” category of stochastic processes (between independent sequences and weakly stationary sequences)?
- ▶ There are dependencies in financial and economic data series, so the independence assumption is often not warranted.
- ▶ Much of statistical analysis involving confidence intervals and hypothesis testing relies on asymptotic normality.
- ▶ We need to have processes for which there is a central limit theorem.
- ▶ Given $\{(X_t, \mathcal{F}_t), t \geq 0\}$ is a MDS with common finite variance σ^2 , under some additional conditions

$$\sqrt{T}\bar{X} \Rightarrow N(0, \sigma^2).$$

Wold Decomposition: 1

- ▶ A very important theorem by Wold describes the structure of weakly stationary processes. This is referred to as the Wold decomposition.
- ▶ This decomposition begins with a general weakly stationary process $\{X_t, -\infty < t < +\infty\}$.
- ▶ We define $\mathcal{F}_t$ to be the σ -algebra generated by $\{X_s, -\infty < s \leq t\}$.
- ▶ Suppose $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\Phi, \Omega\}$.
- ▶ Then the $\{X_t\}$ process can be written as a **moving average process of infinite order** with respect to some white noise process $\{\epsilon_t, -\infty < t < \infty\}$. That is

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j},$$

where $\psi_0 = 1$ and $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

Wold Decomposition: 2

- ▶ The process is linear and causal and is represented as a moving average of infinite order with respect to a white noise process.
- ▶ Of course there are infinitely many parameters, so we must look for a simpler model that can be estimated with a finite data set, e.g. a moving average with q lags (MA(q)), an autoregressive process with p lags (AR(p)), or a combination of both, ARMA(p, q).
- ▶ Note, the condition $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\Phi, \Omega\}$ is restrictive and there are cases where this is false. In this case, the Wold decomposition has two components: the moving average of infinite order plus a “deterministic process.” We give an example of this next.

Wold Decomposition: 3

- ▶ Suppose we have two uncorrelated random variables, U and V with $E(U) = E(V) = 0$ and $Var(U) = Var(V) = \sigma^2$.
Let

$$X_t = U \cos(\theta t) + V \sin(\theta t).$$

- ▶ $E(X_t) = 0$ and $Var(X_t) = \sigma^2(\cos^2(\theta t) + \sin^2(\theta t)) = \sigma^2, \forall t$

$$\begin{aligned} Cov(X_{t_1}, X_{t_2}) &= Cov(U \cos(\theta t_1) + V \sin(\theta t_1), U \cos(\theta t_2) + V \sin(\theta t_2)) \\ &= \sigma^2(\cos(\theta t_1) \cos(\theta t_2) + \sin(\theta t_1) \sin(\theta t_2)) \\ &= \sigma^2 \cos(\theta(t_2 - t_1)). \end{aligned}$$

So $\gamma(h) = \sigma^2 \cos(\theta h)$ and $\rho(h) = \cos(\theta h)$

Wold Decomposition: 4

- ▶ Thus the $\{X_t\}$ process is weakly stationary.
- ▶ However, $\mathcal{F}_s = \sigma(U, V)$, $\forall s$ so $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \sigma(U, V)$.
- ▶ This process is perfectly predictable. Given $\mathcal{F}_s$ for any s , the rest of the process is deterministic.
- ▶ This process is called “deterministic” even though it is created by two random variables U and V .
- ▶ Leaving aside the possibility of predictable components, how can we construct a white noise process, $\{\epsilon_t, -\infty < t < \infty\}$, and a set of constants, $\{\psi_j\}$, so that

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$$

Wold Decomposition: 5

Define $\{\epsilon_t, -\infty < t < \infty\}$ by

$$\epsilon_t = X_t - E(X_t | \mathcal{F}_{t-1}).$$

Define $\{\psi_t, 0 < t < \infty\}$ by

$$\psi_i = \gamma_X(i) / \sigma^2 = \rho_X(i).$$

It follows that $\{\epsilon_t, -\infty < t < \infty\}$ is a white noise process (see Course Notes, pp 16-19). Finally,

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$$

Wold Decomposition: 6

- ▶ Under the vanishing σ -algebra condition, we find that all weakly stationary processes can be represented as a **linear, causal** process.
- ▶ A moving average process of infinite order.
- ▶ Of course, all datasets are finite, and we cannot fit processes of infinite order. Thus we need to develop a family of simpler models with a small number of parameters.
- ▶ Those basic models include autoregressive models of order p , $AR(p)$, moving average models of order q , $MA(q)$, and autoregressive moving average models of order (p,q) , $ARMA(p,q)$.

AUTOREGRESSIVE PROCESSES

$$X_t = \phi_0 + \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t,$$

where $\{\epsilon_t\}$ is $WN(0, \sigma^2)$.

Time Series Ideas: AR(1) Processes: 1

- ▶ Consider an autoregressive process of order 1, namely:

$$X_t = \alpha + \beta X_{t-1} + \epsilon_t,$$

where $\{\epsilon_t\}$ is a stationary process with

$$E(\epsilon_t) = 0,$$

$$\text{Var}(\epsilon_t) = \sigma_\epsilon^2.$$

- ▶ One can find $E(X_t)$ and $\text{Var}(X_t)$ as solutions to simple difference equations, which can be iterated back to time 0:
- ▶ $E(X_t) = \alpha + \beta E(X_{t-1})$
- ▶ $\text{Var}(X_t) = \beta^2 \text{Var}(X_{t-1}) + \sigma_\epsilon^2$

Time Series Ideas: AR(1) Processes: 2

Assuming the process is stationary, hence with common mean μ and variance σ^2 ,

► $\mu = \alpha + \beta\mu$, or

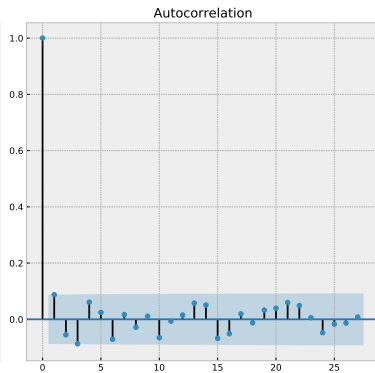
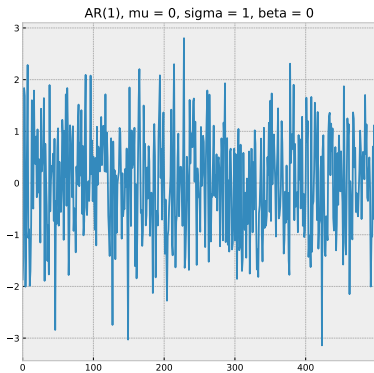
$$\mu = \alpha/(1 - \beta).$$

► $\sigma_X^2 = \beta^2\sigma_X^2 + \sigma_\epsilon^2$ or

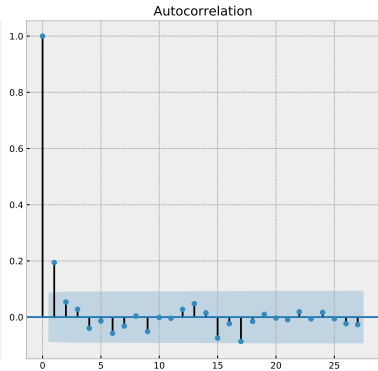
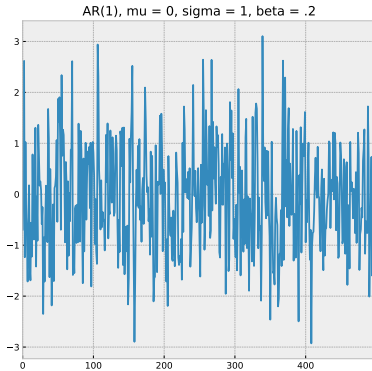
$$\sigma_X^2 = \sigma_\epsilon^2/(1 - \beta^2).$$

- Both require $|\beta| < 1$. Backward iteration of the AR(1) equation will show the necessity and sufficiency of this condition.

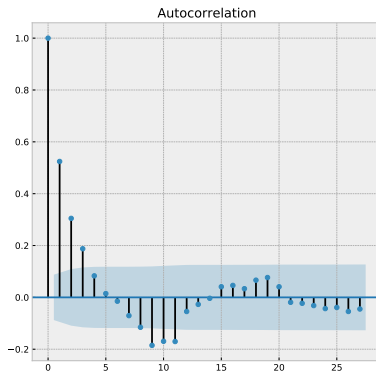
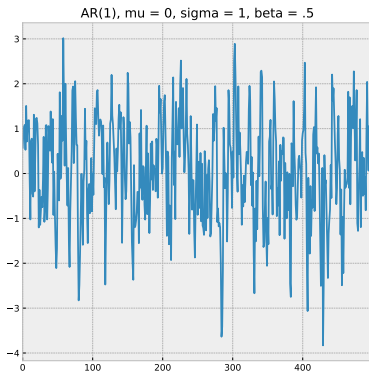
AR(1), $\beta = 0$



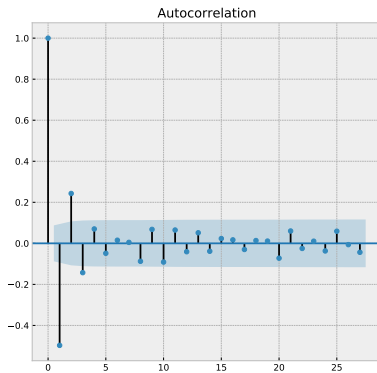
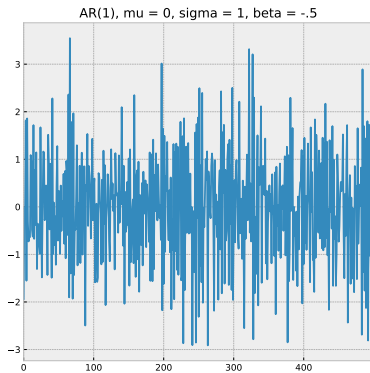
AR(1), $\beta = .2$



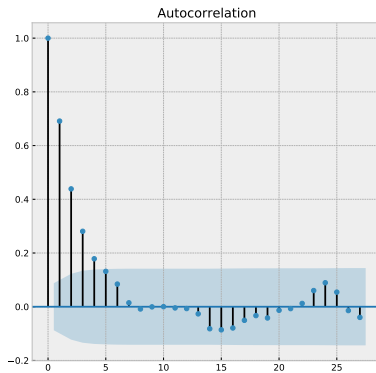
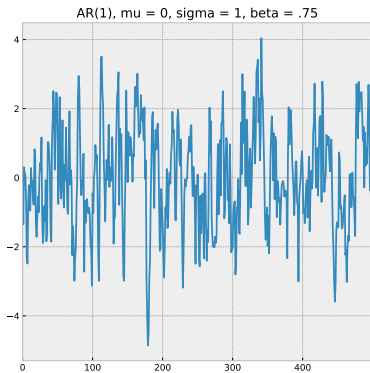
AR(1), $\beta = .5$



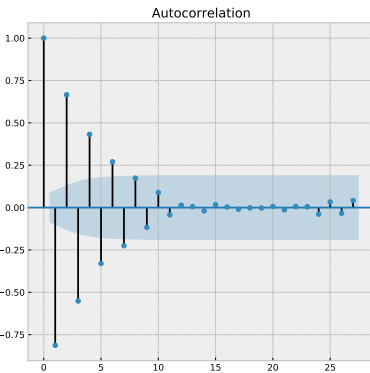
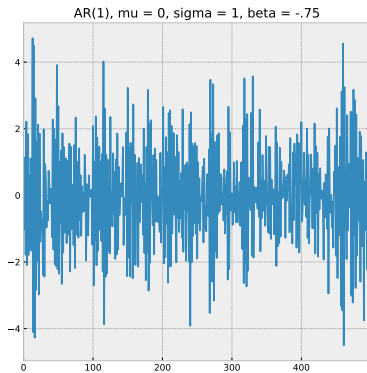
AR(1), $\beta = -.5$



AR(1), $\beta = .75$



AR(1), $\beta = -.75$



AR(1), $\beta = 1$

